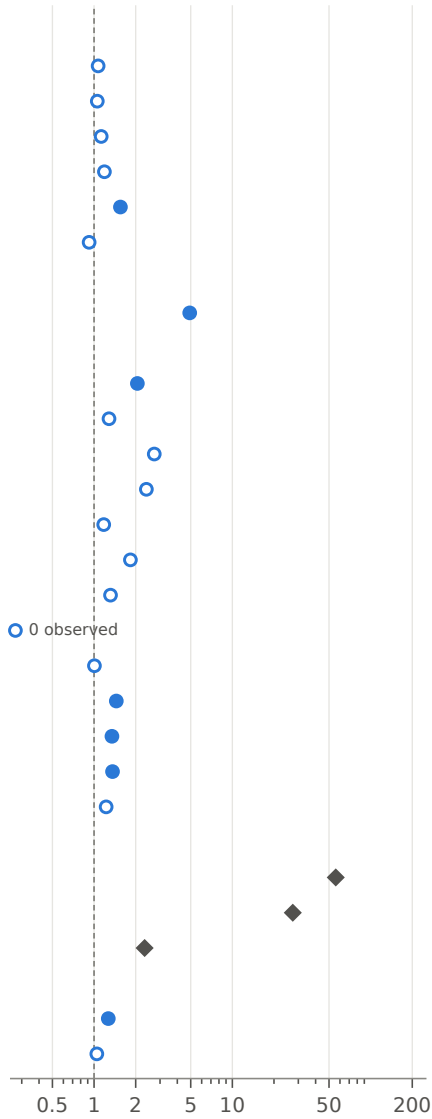
A

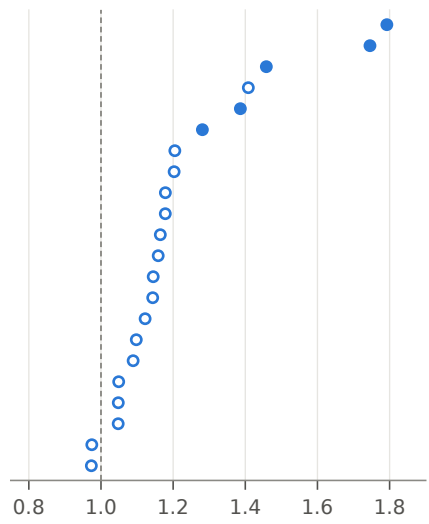


	obs / exp loci	p, matched null loci	MAGMA p (gene level)
Mendelian disease genes			
Any Mendelian (OMIM or Orphanet)	123 / 114.9	0.16	0.75
OMIM	118 / 111.8	0.22	0.74
Orphanet (assessed)	111 / 98.5	0.054	0.92
ClinGen moderate–definitive	77 / 64.8	0.041	0.39
Autosomal dominant	80 / 51.6	1.1×10^{-5}	0.28
Autosomal recessive	74 / 80.3	0.84	0.99
Positive control			
GO: circadian rhythm	22 / 4.5	8.4×10^{-10}	0.0019
Pre-specified phenotypes			
Sleep disturbance	34 / 16.5	3.5×10^{-5}	0.031
without neurodevelopmental features	4 / 3.1	0.38	0.19
Sleep–wake cycle / melatonin	3 / 1.1	0.098	0.23
Prolonged QT interval	3 / 1.3	0.13	0.86
Arrhythmia	17 / 14.5	0.28	0.84
Hypertrophic cardiomyopathy	13 / 7.1	0.024	0.88
Cardiomyopathy	21 / 16.0	0.11	0.58
Reduced circulating complement	0 / 0.90	1.00	0.60
Immunodeficiency	5 / 5.0	0.56	0.99
Global developmental delay	76 / 52.5	1.7×10^{-4}	0.17
Intellectual disability	71 / 52.7	0.0029	0.089
Seizure	70 / 51.5	0.0022	0.57
Short stature	44 / 36.0	0.082	0.22
Post hoc			
Monoamine neurotransmitter disorders	4 / 0.07	6.7×10^{-7}	0.032
Decreased CSF homovanillic acid	3 / 0.11	1.5×10^{-4}	0.092
Complement deficiency genes (IUIS)	2 / 0.86	0.21	0.52
Rhythmic expression			
Rhythmic in ≥3 mouse tissues	105 / 82.8	0.0014	0.0086
Human day–night (GTEx)	44 / 42.0	0.38	0.011

Null loci matched on MAF, LD, gene count and distance

● FDR q < 0.05 within set family ○ FDR q ≥ 0.05 ◆ post hoc set (q not computed)

B



	obs / exp loci	p, matched null loci	MAGMA p (gene level)
Voice	20 / 11.2	0.0078	0.47
Breast	21 / 12.0	0.0092	0.68
Endocrine system	50 / 34.3	0.0029	0.79
Prenatal development or birth	36 / 25.6	0.020	0.60
Limbs	82 / 59.1	4.4×10^{-4}	0.72
Genitourinary system	80 / 62.4	0.0050	0.79
Head or neck	94 / 78.0	0.015	0.72
Digestive system	82 / 68.2	0.025	0.63
Neoplasm	27 / 22.9	0.21	0.99
Musculoskeletal system	102 / 86.6	0.020	0.73
Ear	70 / 60.1	0.076	0.77
Growth abnormality	72 / 62.1	0.077	0.94
Cardiovascular system	76 / 66.4	0.087	0.95
Nervous system	106 / 92.7	0.042	0.57
Integument	68 / 60.6	0.14	0.95
Metabolism/homeostasis	68 / 61.9	0.19	0.92
Eye	83 / 76.2	0.18	0.58
Immune system	49 / 46.7	0.37	0.78
Constitutional symptom	32 / 30.5	0.42	0.99
Respiratory system	54 / 51.5	0.37	0.72
Abnormal cellular phenotype	18 / 18.5	0.59	0.95
Blood and blood-forming tissues	36 / 37.0	0.60	0.94